

Greedy expansions

This document is about summarizing everything I know about greedy expansions, with the aim of exporting them easily to papers/appendices. We do everything for the general case $\beta \in (1, \infty)$.

In all what follows, $\mathbb{N}^{\mathbb{N}}$ and its subsets are endowed with the lexicographical ordering, $\mathbb{R}$ and its subsets are endowed with the natural ordering. We define $\mathbf{I} := [0, 1]$, and $\{\cdot\} : \mathbb{R} \rightarrow [0, 1]$ is the fractional part.

1 Two equivalent definitions

1.1 Rényi's definition

Historically, the *greedy β -expansion* was introduced by A. Rényi in [6, Section 4], under the simpler name of β -expansion. The definition relies on a map that, following [2, Abstract], we call β -transformation.

Definition 1.1. (*β -transformation*) Let $\beta \in (1, \infty)$. We define the β -transformation $T_\beta : \mathbf{I} \rightarrow [0, 1]$ by

$$T_\beta(x) = \{\beta x\}, \quad \forall x \in \mathbf{I}. \quad (1)$$

We also define the β -digitization $d_\beta : \mathbf{I} \rightarrow \mathbb{N}$ by

$$d_\beta(x) = \lfloor \beta x \rfloor, \quad \forall x \in \mathbf{I}. \quad (2)$$

Note that $T_\beta(x) = \beta x - d_\beta(x)$, $\forall x \in \mathbf{I}$.

Rényi defined the greedy β -expansion to be the successive digits generated from iterations of the β -transformations.

Definition 1.2. (*Rényi's greedy β -expansion*) Let $\beta \in (1, \infty)$. Then, we define the Rényi's greedy β -expansion $\mathbf{g}_\beta : \mathbf{I} \rightarrow \mathbb{N}^{\mathbb{N}}$ by

$$\mathbf{g}_\beta(x) = \prod_{i=0}^{\infty} d_\beta(T_\beta^i(x)), \quad \forall x \in \mathbf{I}. \quad (3)$$

For $x \in \mathbf{I}$, we say that $\mathbf{g}_\beta(x)$ is the Rényi's greedy β -expansion of x .

The name greedy β -expansion of x is justified because $\mathbf{g}_\beta(x)$ is exactly analogous to the usual definition of the expansion of x in some integer base, as the next lemma shows.

Lemma 1.1. [6, Section 1] (*Rényi's greedy expansion represents real numbers*) Let $\beta \in (1, \infty)$, and $x \in \mathbf{I}$. Then,

$$x = \sum_{i=1}^{\infty} \mathbf{g}_\beta(x)_i \beta^{-i}. \quad (4)$$

Lemma 1.1 is infact a direct corollary of the following stronger statement, that quantifies how fast the Renyi's greedy β -expansion approximates the real numbers.

Lemma 1.2. (*Approximation rate of Renyi's greedy β -expansion*) Let $\beta \in (1, \infty)$, and $x \in \mathbf{I}$. Then,

$$x = \sum_{i=1}^n \mathbf{g}_\beta(x)_i \beta^{-i} + T_\beta^n(x) \beta^{-n}, \quad \forall n \in \mathbb{N}. \quad (5)$$

Concretely, the partial sum $\sum_{i=1}^n \mathbf{g}_\beta(x)_i \beta^{-i}$ approximates x with error $T_\beta^n(x) \beta^{-n}$.

Proof. Let $\beta \in (1, \infty)$, $x \in \mathbf{I}$ and $n \in \mathbb{N}$. Note that

$$\frac{1}{\beta^n} T_\beta^n(x) = \frac{1}{\beta^n} T_\beta(T_\beta^{n-1}(x)) = \frac{1}{\beta^n} \{\beta T_\beta^{n-1}(x)\} \quad (6)$$

$$= \frac{1}{\beta^n} \beta T_\beta^{n-1}(x) - \frac{1}{\beta^n} d_\beta(T_\beta^{n-1}(x)) \quad (7)$$

$$= \frac{1}{\beta^{n-1}} T_\beta^{n-1}(x) - \frac{1}{\beta^n} \mathbf{g}_\beta(x)_n. \quad (8)$$

By induction on n , we get

$$\frac{1}{\beta^n} T_\beta^n(x) = \frac{1}{\beta^0} T_\beta^0(x) - \sum_{i=1}^n \frac{1}{\beta^i} \mathbf{g}_\beta(x)_i = x - \sum_{i=1}^n \frac{1}{\beta^i} \mathbf{g}_\beta(x)_i. \quad (9)$$

□

1.2 Erdős's definition

33 years later, P. Erdős defined in [4] the greedy β -expansion to be the lexicographically largest β -expansion, with the definition of β -expansion being quite different from the one of A. Rényi.

Definition 1.3 (β -expansion). *Let $\beta \in (1, \infty)$. We say that $p \in \mathbb{N}^{\mathbb{N}}$ is a β -expansion of $x \in \mathbf{I}$ if*

$$x = \sum_{i=1}^{\infty} p_i \beta^{-i} \text{ and } p_i \in \{0, \dots, \lceil \beta \rceil - 1\} \quad \forall i \in \mathbb{N}. \quad (10)$$

We can reformulate the above equation. Let $\delta_\beta : \subseteq \mathbb{N}^{\mathbb{N}} \rightarrow \mathbb{R}$ be such that

$$\Omega_\beta := \text{dom}(\delta_\beta) := \{0, \dots, \lceil \beta \rceil - 1\}^{\mathbb{N}} \quad (11)$$

and

$$\delta_\beta(p) = \sum_{i=1}^{\infty} p_i \beta^{-i}, \quad \forall p \in \Omega_\beta. \quad (12)$$

Then, Equation (10) can be rewritten

$$x = \delta_\beta(p). \quad (13)$$

For $x \in \mathbf{I}$, we denote the *set of β -expansions of x* by

$$\Omega_\beta(x) := \{p \in \Omega_\beta : x = \delta_\beta(p)\} \subset \Omega_\beta. \quad (14)$$

From this definition, we build-up to define the greedy β -expansion according to Erdős.

Definition 1.4. (*Erdős's greedy β -expansion*) *Let $\beta \in (1, \infty)$. We call Erdős's greedy β -expansion the function $x \mapsto \text{lexsup } \Omega_\beta(x)$, and we say that $\text{lexsup } \Omega_\beta(x)$ is the Erdős's greedy β -expansion of x , for $x \in \mathbf{I}$.*

We can prove that actually, both definitions agree, as remarked originally by P. Erdős and collaborators, because they use the two definitions interchangeably in [3] and [4].

Theorem 1.3. (*Rényi=Erdős greedy β -expansion*)

$$\mathbf{g}_\beta(x) = \text{lexsup } \Omega_\beta(x), \quad \forall x \in \mathbf{I}. \quad (15)$$

Proof. Let $\beta \in (1, \infty)$ and $x \in \mathbf{I}$. First, Lemma 1.1 asserts that $\mathbf{g}_\beta(x) \in \Omega_\beta(x)$. Suppose that $\mathbf{g}_\beta(x) \neq \text{sup } \Omega_\beta(x)$. Then, there exists $p \in \Omega_\beta(x)$ such that $p >_L \mathbf{g}_\beta(x)$, i.e., there exists $i \in \mathbb{N}$ such that $p_j = \mathbf{g}_\beta(x)_j$ for all $j < i$, and $p_i > \mathbf{g}_\beta(x)_i =: k$. Note that $k = d_\beta(T_\beta^{i-1}(x)) = \lfloor \beta T_\beta^{i-1}(x) \rfloor$, i.e. $T_\beta^{i-1}(x) \in [k/\beta, (k+1)/\beta)$. By Equation (5), one gets

$$(k+1)\beta^{-i} > T_\beta^{i-1}(x)\beta^{-i+1} = x - \delta_\beta(\mathbf{g}_\beta(x)|_{i-1}) = x - \delta_\beta(p|_{i-1}). \quad (16)$$

Then,

$$\delta_\beta(p) \geq \delta_\beta(p|_i) = \delta_\beta(p|_{i-1}) + \beta^{-i}p_i \geq \delta_\beta(p|_{i-1}) + \beta^{-i}(k+1) \stackrel{(16)}{>} x, \quad (17)$$

i.e., $p \notin \Omega_\beta(x)$, which is a contradiction. $\square$

From now on, we simply call $\mathbf{g}_\beta$ the *greedy β -expansion*, and for $x \in \mathbf{I}$, we say that $\mathbf{g}_\beta(x)$ is the *greedy β -expansion of x* .

1.3 The set of β -prefixes and β -adic numbers

In this part, we introduce and study β -adic numbers. We say that $p \in \{0, \dots, \lceil \beta \rceil - 1\}^{\mathbb{N}}$ is *finite* if there exists $n \in \mathbb{N}$ such that $p_i = 0$ for all $i > n$. We say that $x \in \mathbf{I}$ is a *β -adic number* (pronounced “betadic number”) if it has a finite β -expansion. We denote by $\mathbb{Q}_\beta \subseteq \mathbf{I}$ the set β -adic numbers. Note that by virtually extending δ_β to finite sequences, through the notation

$$\delta_\beta(w) := \delta_\beta(w0^\infty), \quad \forall w \in \{0, \dots, \lceil \beta \rceil - 1\}^*, \quad (18)$$

one gets

$$\mathbb{Q}_\beta = \delta_\beta(\{0, \dots, \lceil \beta \rceil - 1\}^*) \cap \mathbf{I}. \quad (19)$$

The above form of the set of β -adic numbers motivates the definition of the *set of n -finite β -adic numbers* as

$$\mathbb{Q}_{\beta,n} = \delta_{\beta}(\{0, \dots, \lceil \beta \rceil - 1\}^n) \cap \mathbf{I}, \quad \forall n \in \mathbb{N}. \quad (20)$$

Note that for $p \in \mathbb{N}$, $\mathbb{Q}_p$ is the usual real subfield of p -adic numbers. We further emphasize that for $\beta \notin \mathbb{N}$, $\mathbb{Q}_{\beta}$ is not necessarily a subfield nor a subring of $\mathbb{R}$. See FROUGNY for related work.

Since every $x \in \mathbf{I}$ has a β -expansion, the closure of $\mathbb{Q}_{\beta}$ is $\mathbf{I}$. Therefore, the sets $\mathbb{Q}_{\beta,n}$ can be interpreted as increasingly precise approximations of $\mathbf{I}$. This motivates the definition a local equivalent of $\mathbb{Q}_{\beta,n}$, i.e., a set of β -adic numbers that approximate a specific $x \in \mathbf{I}$. This definition is in two steps: first, we introduce a set of finite sequences that approximate β -expansions of x , and then we use these sequences to build the β -adic numbers that approximate x . For $n \in \mathbb{N}$ and $x \in \mathbf{I}$, we define the *set of n -long β -prefixes of x* as

$$W_{\beta,n}(x) := \{p|_n : p \in \Omega_{\beta}(x)\}, \quad (21)$$

and the *set of n -finite β -adic approximates of x* as

$$\mathbb{Q}_{\beta,n}(x) := \delta_{\beta}(W_{\beta,n}(x)). \quad (22)$$

We now give a characterization of $\mathbb{Q}_{\beta,n}(x)$.

Lemma 1.4. (*Characterization of the set of n -finite β -adic approximates of x*) Let $\beta \in (1, \infty)$, $n \in \mathbb{N}$ and $x \in \mathbf{I}$. Then,

$$\mathbb{Q}_{\beta,n}(x) = \mathbb{Q}_{\beta,n} \cap \left[x - \frac{\lceil \beta \rceil \beta^{-n}}{\beta - 1}, x \right]. \quad (23)$$

Proof. □

2 The set of greedy β -expansions

We first introduce $\widehat{\Omega}_{\beta}$ to be the *set of greedy β -expansions*, usually called *β -shift* [7], which in mathematical symbols is expressed as

$$\widehat{\Omega}_{\beta} := \text{range}(\mathbf{g}_{\beta}) = \{\mathbf{g}_{\beta}(x) : x \in \mathbf{I}\} \subseteq \Omega_{\beta}. \quad (24)$$

From now on, we will study the behavior of $\mathbf{g}_{\beta}$ considered as a function from $\mathbf{I}$ to $\widehat{\Omega}_{\beta}$. A first evident property is that $\mathbf{g}_{\beta}$ is bijective.

Lemma 2.1. (*Greedy β -expansion is bijective*) $\mathbf{g}_{\beta} : \mathbf{I} \rightarrow \widehat{\Omega}_{\beta}$ is bijective, of inverse $\delta_{\beta}|_{\widehat{\Omega}_{\beta}}$.

Proof. By Lemma 1.1, $\mathbf{g}_{\beta}$ is injective and

$$\delta_{\beta}(\mathbf{g}_{\beta}(x)) = x, \quad \forall x \in \mathbf{I}. \quad (25)$$

This makes $\mathbf{g}_{\beta}$ bijective onto its range, with inverse $\delta_{\beta}|_{\widehat{\Omega}_{\beta}}$. □

A following very important property is that $\mathbf{g}_{\beta} : (\mathbf{I}, <) \rightarrow (\widehat{\Omega}_{\beta}, <_L)$ is an order isomorphism, i.e. a order-preserving bijection with order-preserving inverse.

Lemma 2.2. (*Greedy β -expansion preserves order*) [5, Lemma 3] Let $x, t \in \mathbf{I}$, and $\beta \in (1, \infty)$. Then,

$$x < t \iff \mathbf{g}_{\beta}(x) <_L \mathbf{g}_{\beta}(t), \quad (26)$$

i.e. both $\mathbf{g}_{\beta}$ and $\delta_{\beta}|_{\widehat{\Omega}_{\beta}}$ are order-preserving.

Proof. Let $x, t \in \mathbf{I}$, and $\beta \in (1, \infty)$. First note that, by bijectivity of $\mathbf{g}_{\beta}$ (Lemma 2.1), $x \neq t \iff \mathbf{g}_{\beta}(x) \neq \mathbf{g}_{\beta}(t)$. It then suffices to prove that (a) $x < t \Rightarrow \mathbf{g}_{\beta}(x) \leq_L \mathbf{g}_{\beta}(t)$ and (b) $\mathbf{g}_{\beta}(x) <_L \mathbf{g}_{\beta}(t) \Rightarrow x \leq t$. Note that (b) is simply the contraposition of (a), so we only need to prove (a). We proceed by contradiction: suppose that $x < t$, and that $\mathbf{g}_{\beta}(x) >_L \mathbf{g}_{\beta}(t)$, i.e., that there exists $i \in \mathbb{N}$ such that $\mathbf{g}_{\beta}(x)_j = \mathbf{g}_{\beta}(t)_j$, for all $j < i$, and $\mathbf{g}_{\beta}(x)_i > \mathbf{g}_{\beta}(t)_i$. Hence,

$$t > x \geq \sum_{j=1}^i \mathbf{g}_{\beta}(x)_j \beta^{-j} > \sum_{j=1}^i \mathbf{g}_{\beta}(t)_j \beta^{-j} \geq t - \beta^{-i}. \quad (27)$$

So $\mathbf{g}_{\beta}(x)|_i$ is the i -prefix of some β -expansion p of t , satisfying $p >_L \mathbf{g}_{\beta}(t)$, which is a contradiction. □

2.1 Greedy β -prefixes

We further study $\widehat{\Omega}_\beta$ by studying the properties of the finite prefixes of its elements. For $n \in \mathbb{N}$, we define the *set of greedy β -prefixes of length n* defined as

$$\widehat{W}_{\beta,n} := \{p|_n : p \in \widehat{\Omega}_\beta\}. \quad (28)$$

Note that $\widehat{W}_{\beta,n}$ is a finite set of cardinality $N_\beta(n) := \#(\widehat{W}_{\beta,n})$. Note that since $\widehat{W}_{\beta,n} \subseteq \{0, \dots, \lceil \beta \rceil - 1\}$, $N_\beta(n) \leq \lceil \beta \rceil^n$. We denote each element of $\widehat{W}_{\beta,n}$ by $w_{\beta,n}(i)$, for $i \in \{0, \dots, N_\beta(n) - 1\}$, such that

$$w_{\beta,n}(0) <_L w_{\beta,n}(1) <_L \dots <_L w_{\beta,n}(N_\beta(n) - 1). \quad (29)$$

We finally introduce the *set of greedy β -prefixes* (of arbitrary length) as

$$\widehat{W}_\beta := \bigcup_{n=0}^{\infty} \widehat{W}_{\beta,n}. \quad (30)$$

An important property of greedy β -prefixes is that they characterize the complete greedy β -expansion of some numbers.

Lemma 2.3. (*Greedy β -prefixes correspond to greedy β -expansions*) *Let $\beta \in (1, \infty)$. Then,*

$$\mathbf{g}_\beta(\delta_\beta(w)) = w0^\infty, \quad \forall w \in \widehat{W}_\beta. \quad (31)$$

Proof. Let $\beta \in (1, \infty)$, $w \in \widehat{W}_\beta$, and $n := |w|$. By means of contradiction, suppose that $w0^\infty$ is not the greedy β -expansion of $x := \delta_\beta(w)$, which by Theorem 1.3 yields $q := \mathbf{g}_\beta(\delta_\beta(w)) >_L w0^\infty =: p$, i.e., there exists $i \in \mathbb{N}$ such that $p_j = p_j$, for all $j < i$, and $1 = p_i > p_i = 0$.

(a) Suppose that $i > n$. Then,

$$\delta_\beta(q) \geq \delta_\beta(q_{1:n}) + \beta^{-i} p_i \quad (32)$$

$$= \delta_\beta(w) + \beta^{-i} p_i \quad (33)$$

$$> \delta_\beta(w) = x, \quad (34)$$

so q is not a β -expansion of $\delta_\beta(w)$, which is a contradiction to $\text{eval}(q) = x$.

(b) Suppose that $i \leq n$. Since $w \in \widehat{W}_{\beta,n}$, there exists $x \in \mathbf{I}$ such that $\mathbf{g}_\beta(x)|_n = w$. Note that

$$x = \delta_\beta(\mathbf{g}_\beta(x)) \geq \delta_\beta(\mathbf{g}_\beta(x)|_n) = \delta_\beta(w), \quad (35)$$

and that

$$\mathbf{g}_\beta(x)|_{i-1} = w|_{i-1} = q|_{i-1} = \mathbf{g}_\beta(\delta_\beta(w))|_{i-1}. \quad (36)$$

Hence,

$$T_\beta^{i-1}(x) = \beta^{i-1}(x - \delta_\beta(\mathbf{g}_\beta(x)_{i-1})) \quad (37)$$

$$\geq \beta^{i-1}(\delta_\beta(w) - \delta_\beta(\mathbf{g}_\beta(x)_{i-1})) \quad (38)$$

$$= \beta^{i-1}(\delta_\beta(w) - \delta_\beta(\mathbf{g}_\beta(\delta_\beta(w))_{i-1})) \quad (39)$$

$$= T_\beta^{i-1}(\delta_\beta(w)). \quad (40)$$

However, $\mathbf{g}_\beta(\delta_\beta(w))_i = p_i = 1$, so $T_\beta^{i-1}(\delta_\beta(w)) > 1/\beta$, and $\mathbf{g}_\beta(x)_i = w_i = 0$, so $T_\beta^{i-1}(x) \leq 1/\beta$, which yields a contradiction. □

We immediately deduce the important stability under 0-suffixing of the greedy β -prefixes, mentioned first in the proof of [6, Theorem 2].

Corollary 2.4. (*Stability of the set of greedy β -prefixes under 0-suffixing*)

$$w \in \widehat{W}_\beta \iff w0 \in \widehat{W}_\beta \iff w0^\infty \in \widehat{\Omega}_\beta. \quad (41)$$

2.2 Greedy β -adic numbers

We first introduce a new terminology element. We say that $p \in \{0, \dots, \lceil \beta \rceil - 1\}^{\mathbb{N}}$ is *finite* if there exists $n \in \mathbb{N}$ such that $p_i = 0$ for all $i > n$. We say that $x \in \mathbf{I}$ is a *β -adic number* (pronounced “betadic number”) if it has a finite β -expansion, and that $x \in \mathbf{I}$ is a *greedy β -adic number* if its greedy β -expansion is finite. We denote by $\widehat{\mathbb{Q}}_{\beta}$ the set of greedy β -adic numbers in $\mathbf{I}$, i.e.

$$\widehat{\mathbb{Q}}_{\beta} := \delta_{\beta}(\widehat{W}_{\beta}). \quad (42)$$

Moreover, we also define the set of *greedy β -adic numbers of order n* to be the greedy β -adic numbers defined only by the prefix of length n of their greedy β -expansion, i.e.,

$$\widehat{\mathbb{Q}}_{\beta,n} := \delta_{\beta}(\widehat{W}_{\beta,n}). \quad (43)$$

Note that

$$\widehat{\mathbb{Q}}_{\beta} = \bigcup_{n=0}^{\infty} \widehat{\mathbb{Q}}_{\beta,n}. \quad (44)$$

Sometimes it will be more convenient to use

$$\widehat{\mathbb{Q}}_{\beta,n}^+ = \widehat{\mathbb{Q}}_{\beta,n} \cup \{1\}. \quad (45)$$

We denote the elements of $\widehat{\mathbb{Q}}_{\beta,n}^+$ by $x_{\beta,n}(i)$, for $i \in \{0, \dots, N_{\beta}(n)\}$, such that

$$0 = x_{\beta,n}(0) < x_{\beta,n}(1) < \dots < x_{\beta,n}(N_{\beta}(n) - 1) \leq x_{\beta,n}(N_{\beta}(n)) = 1, \quad (46)$$

Since by Lemma 2.2, $\mathbf{g}_{\beta}$ preserves order, this yields the following Corollary.

Corollary 2.5.

$$x_{\beta,n}(i) = \delta_{\beta}(w_{\beta,n}(i)) \text{ and } w_{\beta,n}(i) = \mathbf{g}_{\beta}(x_{\beta,n}(i))|_n, \quad \forall i \in \{0, \dots, N_{\beta}(n) - 1\}. \quad (47)$$

Note that since $\widehat{\Omega}_{\beta}$ is the closure of $\widehat{W}_{\beta}0^{\infty}$ in the Cantor space topology, then $\mathbf{I}$ is the closure of $\widehat{\mathbb{Q}}_{\beta}$ in the Euclidean topology. We first derive a simple characterization of the elements of $\widehat{\mathbb{Q}}_{\beta,n}$.

Lemma 2.6. (*Characterization of greedy β -adic numbers*) Let $\beta \in (1, \infty)$ and $n \in \mathbb{N}$. Then,

$$x \in \widehat{\mathbb{Q}}_{\beta,n} \iff x = \delta_{\beta}(\mathbf{g}_{\beta}(x)|_n). \quad (48)$$

Proof. First, recall that by (5), $x \leq \delta_{\beta}(\mathbf{g}_{\beta}(x)|_n)$, for all $x \in \mathbf{I}$. Obviously, if $x = \delta_{\beta}(\mathbf{g}_{\beta}(x)|_n)$, then $x \in \widehat{\mathbb{Q}}_{\beta,n}$. Conversely, suppose that $x > \delta_{\beta}(\mathbf{g}_{\beta}(x)|_n)$ and $x \in \widehat{\mathbb{Q}}_{\beta,n}$, i.e., there exists $w \in \widehat{W}_{\beta,n}$ such that $x = \delta_{\beta}(w)$. Since $x > \delta_{\beta}(\mathbf{g}_{\beta}(x)|_n)$, then by Lemma 2.2, $w >_L \mathbf{g}_{\beta}(x)|_n$, which is in contradiction with Theorem 1.3. $\square$

2.3 Greedy β -expansion as a discrete-optimization problem

In this section, we show that the greedy β -expansion can be reduced to solving a *discrete-optimization problem* [1]: “A discrete-optimization problem is a problem of maximizing a real-valued objective function c on a finite set of feasible solutions S .” We will see that, for $x \in \mathbf{I}$, finding the n first bits of $\mathbf{g}_{\beta}(x)$ correspond to solving a discrete-optimization problem, in which the objective function is the identity, and the finite set S is $\widehat{\mathbb{Q}}_{\beta,n} \cap [0, x]$. This approach will with prove useful to establish further results. We formally introduce the *n -order β -adic solver* as

$$\mathcal{S}_{\beta,n}(x) := \sup(\widehat{\mathbb{Q}}_{\beta,n} \cap [0, x]), \quad \forall x \in \mathbf{I}. \quad (49)$$

Note that $\mathcal{S}_{\beta,n}$ solves a discrete-optimization problem. Alternatively, we also introduce also *n -order β -prefix solver* as

$$\mathcal{S}_{\beta,n}^{\text{lex}}(x) := \text{lexsup}\{w \in \widehat{W}_{\beta,n}, \delta_{\beta}(w) \leq x\}, \quad \forall x \in \mathbf{I}. \quad (50)$$

$\mathcal{S}_{\beta,n}^{\text{lex}}$ also solves a discrete-optimization problem.

Lemma 2.7. (*Greedy β -expansion solves a discrete-optimization problem*) Let $\beta \in (1, \infty)$ and $x \in \mathbf{I}$. Then, for $n \in \mathbb{N}$,

$$\delta_{\beta}(\mathbf{g}_{\beta}(x)|_n) = \mathcal{S}_{\beta,n}(x) \text{ and } \mathbf{g}_{\beta}(x)|_n = \mathcal{S}_{\beta,n}^{\text{lex}}(x). \quad (51)$$

Proof. Let $\beta \in (1, \infty)$, $x \in \mathbf{I}$ and $n \in \mathbb{N}$. First note that, by definition, $\delta_\beta(\mathbf{g}_\beta(x)|_n) \in \widehat{\mathbb{Q}}_{\beta,n}$. Moreover, by Lemma 1.2,

$$\delta_\beta(\mathbf{g}_\beta(x)|_n) \leq \delta_\beta(\mathbf{g}_\beta(x)|_n) + T_\beta^n(x)\beta^{-n} = x. \quad (52)$$

Therefore, $\delta_\beta(\mathbf{g}_\beta(x)|_n) \in \{y \in \widehat{\mathbb{Q}}_\beta : y \leq x\} =: \widehat{\mathbb{Q}}_{\beta,n}(x)$. There remains to prove that $\delta_\beta(\mathbf{g}_\beta(x)|_n) = \sup \widehat{\mathbb{Q}}_{\beta,n}(x) = \mathcal{S}_{\beta,n}(x)$. By sake of contradiction, suppose that

$$\delta_\beta(\mathbf{g}_\beta(x)|_n) < \sup \widehat{\mathbb{Q}}_{\beta,n}(x). \quad (53)$$

$x_{\beta,n}(j) := \delta_\beta(\mathbf{g}_\beta(x)|_n) < \sup\{x_{\beta,n}(i) \in \widehat{\mathbb{Q}}_{\beta,n} : x_{\beta,n}(i) \leq x\} =: x_{\beta,n}(I)$. Then, by Lemma 2.2, $w_{\beta,n}(j) = \mathbf{g}_\beta(x_{\beta,n}(j))|_n <_L \mathbf{g}_\beta(x_{\beta,n}(I))|_n = w_{\beta,n}(I)$. Note that since $x_{\beta,n}(I) \in [x_{\beta,n}(j), x] \subseteq [x - \frac{[\beta]-1}{\beta-1}\beta^{-n}, x]$, $w_{\beta,n}(I)$ is the n -prefix of β -expansion of x . However, since $w_{\beta,n}(I) >_L w_{\beta,n}(j)$, we have a contradiction to the fact that $w_{\beta,n}(j)$ is the a prefix of the greedy β -expansion of x . $\square$

We finally derive an elementary upper bound on the spacings between the consecutive elements of $\widehat{\mathbb{Q}}_{\beta,n}$.

Lemma 2.8. (*Spacings between elements of $\widehat{\mathbb{Q}}_{\beta,n}$*) Let $\beta \in (1, \infty)$, and $n \in \mathbb{N}$. Then, for every $i \in \{0, \dots, N_\beta(n-1) - 1\}$, $x_{\beta,n}(i+1) \leq x_{\beta,n}(i) + \beta^{-n}$.

Proof. Let $\beta \in (1, \infty)$, $n \in \mathbb{N}$, and $r \in [x_{\beta,n}(i), x_{\beta,n}(i+1))$. By Lemma 1.1, we have that

$$r = \delta_\beta(\mathbf{g}_\beta(r)|_n) + T_\beta^n(r)\beta^{-n} \stackrel{\text{Lem. 2.7}}{=} x_{\beta,n}(i) + T_\beta^n(r)\beta^{-n} < x_{\beta,n}(i) + \beta^{-n}. \quad (54)$$

Hence,

$$x_{\beta,n}(i+1) = \lim_{r \rightarrow x_{\beta,n}(i+1)} r \leq \lim_{r \rightarrow x_{\beta,n}(i+1)} x_{\beta,n}(i) + \beta^{-n} = x_{\beta,n}(i) + \beta^{-n}. \quad (55)$$

$\square$

3 Number of greedy expansions

A fundamental question is how big is the set of greedy β -expansions $\widehat{\Omega}_\beta$, which directly relates to the evolution of $N_\beta(n) = \#\widehat{W}_{\beta,n} = \#\widehat{\mathbb{Q}}_{\beta,n}$ for $n \in \mathbb{N}$. We will also study this problem on a local perspective, i.e., estimate the size of the set of all greedy β -expansions of numbers in an interval $I \subseteq \mathbf{I}$.

3.1 The global case

A fundamental question about the nature of $\widehat{W}_{\beta,n}$ (or equivalently $\widehat{\mathbb{Q}}_{\beta,n}$) is its cardinality. The behavior of $N_\beta(n)$ was first established in the proof of [6, Theorem 2]. Note that by Lemma 2.4, we get that $\widehat{\mathbb{Q}}_{\beta,n-1} \subseteq \widehat{\mathbb{Q}}_{\beta,n}$. It will prove useful to consider closely the real numbers in $\widehat{\mathbb{Q}}_{\beta,n}$ that lie between two consecutive numbers in $\widehat{\mathbb{Q}}_{\beta,n-1}$. Accordingly, we define

$$\widehat{\mathbb{Q}}_\beta(n, i) := \widehat{\mathbb{Q}}_{\beta,n} \cap [x_{\beta,n-1}(i), x_{\beta,n-1}(i+1)], \quad (56)$$

and

$$\widehat{\Omega}_\beta(n, i) := \mathbf{g}_\beta(\widehat{\mathbb{Q}}_\beta(n, i))|_n, \quad (57)$$

for all $i \in \{0, \dots, N_\beta(n) - 1\}$.

Lemma 3.1. Let $\beta \in (1, \infty)$, $n \in \mathbb{N}$, and $i \in \{0, \dots, N_\beta(n) - 1\}$. Then,

$$\widehat{\Omega}_\beta(n, i) \subseteq w_{\beta,n}(2 = n-1) \cdot \{0, \dots, [\beta] - 1\}. \quad (58)$$

Proof. Let $\beta \in (1, \infty)$, $n \in \mathbb{N}$, and $i \in \{0, \dots, N_\beta(n) - 1\}$, and suppose that

$$\widehat{\Omega}_\beta(n, i) \not\subseteq w_{\beta,n-1}(i) \{0, \dots, [\beta] - 1\}. \quad (59)$$

Then, there exists $w \in \{0, \dots, [\beta] - 1\}^{n-1}$ and $b \in \{0, \dots, [\beta] - 1\}$, such that $w \neq w_{\beta,n-1}(i)$ and $xb \in \widehat{\Omega}_\beta(n, i)$. Hence, $\delta_\beta(xb) \in \widehat{\mathbb{Q}}_\beta(n, i)$, i.e. $x_{\beta,n-1}(i) = \sup\{x_{\beta,n-1}(j) \in \widehat{\mathbb{Q}}_{\beta,n-1} : x_{\beta,n-1}(j) \leq \delta_\beta(xb)\}$, which implies that

$$w \stackrel{\text{Lem. 2.3}}{=} \mathbf{g}_\beta(\delta_\beta(xb))|_{n-1} \stackrel{\text{Lem. 2.7}}{=} \mathbf{g}_\beta(x_{\beta,n-1}(i))|_{n-1} = w_{\beta,n-1}(i), \quad (60)$$

which yields a contradiction. $\square$

Lemma 3.2. Let $\beta \in (1, \infty)$, $n \in \mathbb{N}$, and $i \in \{0, \dots, N_\beta(n) - 1\}$. Then, there exists $k \in \{0, \dots, \lceil \beta \rceil - 1\}$ such that

$$\widehat{\Omega}_\beta(n, i) = w_{\beta, n-1}(i) \{0, \dots, k\}. \quad (61)$$

Proof. Let $\beta \in (1, \infty)$, $n \in \mathbb{N}$, and $i \in \{0, \dots, N_\beta(n) - 1\}$. First, by previous Lemma, we have that $\widehat{\Omega}_\beta(n, i) \subseteq w_{\beta, n-1}(i) \{0, \dots, \lceil \beta \rceil - 1\}$, i.e., there exists $J \subseteq \{0, \dots, \lceil \beta \rceil - 1\}$, such that $\widehat{\Omega}_\beta(n, i) = w_{\beta, n-1}(i)J$. By Lemma 2.4, $0 \in J$. We let $m := |J|$, $k = \max J$, and define $0 = j_0 < j_1 < \dots < j_{m-1} = k$. We define $\ell \in \{0, \dots, N_\beta(n) - 1\}$ to be such that $w_\ell(n) = w_{\beta, n-1}(i)0$. Then, $w_{\ell+h}(n) = w_{\beta, n-1}(i)j_h$, for $h \in \{2, \dots, m\}$. Suppose that there exists $h \in \{0, \dots, m-2\}$ such that $j_{h+1} \geq j_h + 2$. Then,

$$x_{\beta, n}(\ell + h + 1) = \delta_\beta(w_{\ell+h+1}(n)) = \delta_\beta(w_\ell(n-1)j_{h+1}) = \delta_\beta(w_\ell(n-1)) + j_{h+1}\beta^{-n} \quad (62)$$

$$\geq \delta_\beta(w_\ell(n-1)) + (j_h + 2)\beta^{-n} = \delta_\beta(w_\ell(n)j_h) + 2\beta^{-n} = x_{\beta, n}(\ell + h) + 2\beta^{-n}, \quad (63)$$

which is in contradiction with Lemma 2.8. Therefore, $j_{h+1} = j_h + 1$, for all $h \in \{0, \dots, m-2\}$, i.e. $J = \{0, \dots, k\}$. $\square$

We now introduce a useful transformation, which can be seen as a half-open convex hull of a subset of $\mathbb{R}$, as

$$\text{int}(S) := [\inf S, \sup S), \quad \forall S \subseteq \mathbb{R}. \quad (64)$$

Lemma 3.3. Let $\beta \in (1, \infty)$, $n \in \mathbb{N}$, and $i \in \{0, \dots, N_\beta(n) - 1\}$. Then,

$$|\text{int}(\widehat{\mathbb{Q}}_\beta(n, i))| = \beta^{-n}(\#\widehat{\mathbb{Q}}_\beta(n, i) - 1). \quad (65)$$

Proof. Let $\beta \in (1, \infty)$, $n \in \mathbb{N}$, and $i \in \{0, \dots, N_\beta(n) - 1\}$. By Lemma 3.2, there exists $k \in \{0, \dots, \lceil \beta \rceil - 1\}$ such that $\widehat{\Omega}_\beta(n, i) = w_{\beta, n-1}(i) \{0, \dots, k\}$. Then,

$$\widehat{\mathbb{Q}}_\beta(n, i) = \delta_\beta(\widehat{\Omega}_\beta(n, i)) = \delta_\beta(w_{\beta, n-1}(i)) + \beta^{-n} \{0, \dots, k\} = x_{\beta, n-1}(i) + \beta^{-n} \{0, \dots, k\}. \quad (66)$$

Hence, $\text{int}(\widehat{\mathbb{Q}}_\beta(n, i)) = [x_{\beta, n-1}(i), x_{\beta, n-1}(i) + k\beta^{-n})$, i.e. $|\text{int}(\widehat{\mathbb{Q}}_\beta(n, i))| = k\beta^{-n}$. We conclude the proof by noting that $k = \#(\widehat{\Omega}_\beta(n, i)) - 1 = \#(\widehat{\mathbb{Q}}_\beta(n, i)) - 1$. $\square$

Lemma 3.4. Let $\beta \in (1, \infty)$, $n \in \mathbb{N}$. Then,

$$N_\beta(n) \leq \frac{\beta^{n+1}}{\beta - 1}. \quad (67)$$

Proof.

$$N_\beta(n) = \#\widehat{\mathbb{Q}}_{\beta, n} = \sum_{i=0}^{N_\beta(n-1)-1} \#\widehat{\mathbb{Q}}_\beta(n, i) \quad (68)$$

$$\stackrel{\text{Lem. 3.3}}{=} \sum_{i=0}^{N_\beta(n-1)-1} (\beta^n |\text{int}(\widehat{\mathbb{Q}}_\beta(n, i))| + 1) \quad (69)$$

$$= \beta^n \sum_{i=0}^{N_\beta(n-1)-1} |\text{int}(\widehat{\mathbb{Q}}_\beta(n, i))| + N_\beta(n-1) \quad (70)$$

$$\leq \beta^n + N_\beta(n-1). \quad (71)$$

Hence, since $N_\beta(0) = 1$, we get that

$$N_\beta(n) \leq \sum_{i=0}^n \beta^i = \frac{\beta^{n+1} - 1}{\beta - 1} \leq \frac{\beta^{n+1}}{\beta - 1}. \quad (72)$$

$\square$

Theorem 3.5. (Growth rate of $\#(\widehat{W}_{\beta, n})$) Let $\beta \in (1, \infty)$, and $n \in \mathbb{N}$. Then,

$$\beta^n \leq N_\beta(n) \leq \frac{\beta^{n+1}}{\beta - 1}. \quad (73)$$

Proof. Let $\beta \in (1, \infty)$, and $n \in \mathbb{N}$. The left inequality follows from Lemma 2.8, as

$$1 = x_{\beta, n}(N_\beta(n)) = \sum_{i=0}^{N_\beta(n)-1} (x_{\beta, n}(i+1) - x_{\beta, n}(i)) + x_{\beta, n}(0) = \sum_{i=0}^{N_\beta(n)-1} (x_{\beta, n}(i+1) - x_{\beta, n}(i)) \leq N_\beta(n)\beta^{-n}, \quad (74)$$

which yields $N_\beta(n) \geq \beta^n$. The right inequality is by Lemma 3.4. $\square$

3.2 The local case

In this part, we consider intervals of the form $I = [a, b] \subseteq \mathbf{I}$. We denote by $\mathcal{I}$ the set of such intervals. We study the evolution of $N_\beta(n, I) := \#\widehat{\Omega}_\beta(n, I)$, with $\widehat{\Omega}_\beta(n, I) = \{\mathbf{g}_\beta(x)|_n : x \in I\}$, for $I \in \mathcal{I}$. We also define $\widehat{\mathbb{Q}}_\beta(n, I) := \delta_\beta(\widehat{\Omega}_\beta(n, I))$.

Lemma 3.6. *Let $I \in \mathcal{I}$, $\beta \in (1, \infty)$ and $n \in \mathbb{N}$. Then,*

$$\inf I \in \widehat{\mathbb{Q}}_{\beta, n} \iff \widehat{\mathbb{Q}}_\beta(n, I) = \widehat{\mathbb{Q}}_{\beta, n} \cap I. \quad (75)$$

Proof. Suppose that $\inf I \in \widehat{\mathbb{Q}}_{\beta, n}$. Then, □

Lemma 3.7. *Let $I, J \in \mathcal{I}$, $\beta \in (1, \infty)$ and $n \in \mathbb{N}$. Then,*

$$\widehat{\mathbb{Q}}_\beta(n, I) \subseteq \widehat{\mathbb{Q}}_\beta(n, I \cup J) = \widehat{\mathbb{Q}}_\beta(n, I) \cup \widehat{\mathbb{Q}}_\beta(n, J). \quad (76)$$

Proof. □

We first make assumptions on interval I , and then relax the hypotheses.

Lemma 3.8. *Let $\beta \in (1, \infty)$, and $n \in \mathbb{N}$. Suppose that there exists $m \leq n$ and $i, j \in \{0, \dots, N_\beta(m)\}$, with $i < j$, such that*

$$I = [x_{\beta, m}(i), x_{\beta, m}(j)]. \quad (77)$$

Then,

$$\beta^n |I| \leq N_\beta(n, I) \leq \frac{\beta^{n+1} - \beta^{m+1}}{\beta - 1} |I| + N_\beta(m, I). \quad (78)$$

Proof. Since $x_{\beta, m}(i), x_{\beta, m}(j) \in \widehat{\mathbb{Q}}_\beta(m)$, we also have $x_{\beta, m}(i), x_{\beta, m}(j) \in \widehat{\mathbb{Q}}_{\beta, n}$, i.e., there exist $k, \ell \in \{0, \dots, N_\beta(n) - 1\}$ such that $I = [x_{\beta, n}(k), x_{\beta, n}(\ell)]$. Then, $N_\beta(n, I) = \ell - k$. This allows establish the left side of the inequality, as by Lemma 2.8,

$$|I| = \sup I - \inf I = x_{\beta, n}(\ell) - x_{\beta, n}(k) = \sum_{i=k}^{\ell-1} (x_{\beta, n}(i+1) - x_{\beta, n}(i)) \leq (\ell - k)\beta^{-n} = N_\beta(n, I)\beta^{-n}, \quad (79)$$

which yields $N_\beta(n, I) \geq \beta^n |I|$. We move to the right inequality. Let $p \in \{m+1, \dots, n\}$, and note that by Lemma 2.3, there exists $i_p, j_p \in \{0, \dots, N_\beta(p)\}$ such that $x_{\beta, p}(i_p) = x_{\beta, m}(i)$ and $x_{\beta, p}(j_p) = x_{\beta, m}(j)$, i.e., that $I = [x_{\beta, p}(i_p), x_{\beta, p}(j_p)]$. It follows that

$$\widehat{\mathbb{Q}}_\beta(p, I) = \widehat{\mathbb{Q}}_\beta(p) \cap I = \widehat{\mathbb{Q}}_\beta(p) \cap [x_{\beta, p-1}(i_{p-1}), x_{\beta, p-1}(j_{p-1})] \quad (80)$$

$$= \bigcup_{h=i_{p-1}}^{j_{p-1}-1} \left(\widehat{\mathbb{Q}}_\beta(p) \cap [x_{\beta, p-1}(h), x_{\beta, p-1}(h+1)] \right) \quad (81)$$

$$= \bigcup_{h=i_{p-1}}^{j_{p-1}-1} \widehat{\mathbb{Q}}_\beta(p, h). \quad (82)$$

Then,

$$N_\beta(p, I) = \#\widehat{\mathbb{Q}}_\beta(p, I) = \sum_{h=i_{p-1}}^{j_{p-1}-1} \#\widehat{\mathbb{Q}}_\beta(p, h) \stackrel{\text{Lem. 3.3}}{=} \sum_{h=i_{p-1}}^{j_{p-1}-1} (\beta^p |\text{int}(\widehat{\mathbb{Q}}_\beta(p, h))| + 1) \quad (83)$$

$$= \beta^p \sum_{h=i_{p-1}}^{j_{p-1}-1} |\text{int}(\widehat{\mathbb{Q}}_\beta(p, h))| + (j_{p-1} - i_{p-1}) \quad (84)$$

$$= \beta^p \left| \bigcup_{h=i_{p-1}}^{j_{p-1}-1} \text{int}(\widehat{\mathbb{Q}}_\beta(p, h)) \right| + N_\beta(p-1, I) \quad (85)$$

$$= \beta^p \left| \bigcup_{h=i_{p-1}}^{j_{p-1}-1} [x_{\beta, p-1}(h), x_{\beta, p-1}(h+1)] \right| + N_\beta(p-1, I) \quad (86)$$

$$\leq \beta^p |[x_{\beta,p-1}(i_{p-1}), x_{\beta,p-1}(j_{p-1})]| + N_\beta(p-1, I) \quad (87)$$

$$= \beta^p |I| + N_\beta(p-1, I). \quad (88)$$

By iteration on p spanning from n to $m+1$, we get

$$N_\beta(n, I) \leq \sum_{p=m+1}^n \beta^p |I| + N_\beta(m, I) = \frac{\beta^{n+1} - \beta^{m+1}}{\beta - 1} |I| + N_\beta(m, I). \quad (89)$$

□

We now consider the general case where $I = [a, b]$ with $a, b \in \mathbf{I}$ are arbitrary numbers. We introduce the greedy β -approximation of I of order n by

$$I_\beta(n) = [\delta_\beta(\mathbf{g}_\beta(\inf I)|_n), \delta_\beta(\mathbf{g}_\beta(\sup I)|_n)]. \quad (90)$$

Lemma 3.9. *Let $\beta \in (1, \infty)$ and $n \in \mathbb{N}$.*

$$\widehat{\mathbb{Q}}_\beta(n, I_\beta(n)) \cup \{\delta_\beta(\mathbf{g}_\beta(\sup I)|_n)\} = \widehat{\mathbb{Q}}_\beta(n, I) \cup \{\delta_\beta(\mathbf{g}_\beta(\inf I)|_n)\} = \widehat{\mathbb{Q}}_\beta(n, I_\beta(n) \cup I). \quad (91)$$

Proof. Let $\beta \in (1, \infty)$ and $n \in \mathbb{N}$.

- (a) We prove that $\widehat{\mathbb{Q}}_\beta(n, I_\beta(n)) \cup \{\delta_\beta(\mathbf{g}_\beta(\sup I)|_n)\} = \widehat{\mathbb{Q}}_\beta(n, I_\beta(n) \cup I)$. Remark that $I \setminus I_\beta(n) = [\delta_\beta(\mathbf{g}_\beta(\sup I)|_n), \sup I]$. Then, for all $x \in I \setminus I_\beta(n)$, one has

$$\sup I > x \geq \delta_\beta(\mathbf{g}_\beta(\sup I)|_n) \stackrel{\text{Lem. 2.7}}{=} \sup\{x_{\beta,n}(i) \in \widehat{\mathbb{Q}}_{\beta,n} : x_{\beta,n}(i) \leq \sup I\}, \quad (92)$$

which implies that

$$\delta_\beta(\mathbf{g}_\beta(x)|_n) = \sup\{x_{\beta,n}(i) \in \widehat{\mathbb{Q}}_{\beta,n} : x_{\beta,n}(i) \leq x\} = \sup\{x_{\beta,n}(i) \in \widehat{\mathbb{Q}}_{\beta,n} : x_{\beta,n}(i) \leq \sup I\} = \delta_\beta(\mathbf{g}_\beta(\sup I)|_n). \quad (93)$$

Hence, for all $x \in I \setminus I_\beta(n)$, $x \in \widehat{\mathbb{Q}}_{\beta,n}$ if and only if $x = \delta_\beta(\mathbf{g}_\beta(\sup I)|_n)$, i.e.

□

4 Spacings between greedy β -adic numbers

References

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